

Question	Answer	Marks
3(a)	(gas that obeys) $pV \propto T$ (at all values of p , V and T)	M1
	where T is thermodynamic temperature	A1
3(b)(i)	$pV = \frac{1}{3} Nm\langle c^2 \rangle$ and $Nm / V = \rho$ $(p = \frac{1}{3} \rho \langle c^2 \rangle)$	C1
	r.m.s. speed = $\sqrt{\langle c^2 \rangle}$	C1
	$1.6 \times 10^5 = \frac{1}{3} \times 1.9 \times \langle c^2 \rangle$ leading to r.m.s. speed = 500 m s^{-1} or r.m.s. speed = $\sqrt{(3 \times 1.6 \times 10^5 / 1.9)} = 500 \text{ m s}^{-1}$	A1
3(b)(ii)	$(pV =) \frac{1}{3} Nm\langle c^2 \rangle = NkT$ (so) $\frac{1}{2} m\langle c^2 \rangle = (3 / 2) kT$	C1
	$\frac{1}{2} \times 4.7 \times 10^{-26} \times 503^2 = (3 / 2) \times 1.38 \times 10^{-23} \times T$ $T = 290 \text{ K}$	A1
	or	
	$pV = NkT$ and $Nm / V = \rho$ (so) $T = pm / \rho k$	(C1)
	$T = 1.6 \times 10^5 \times 4.7 \times 10^{-26} / (1.9 \times 1.38 \times 10^{-23})$ $= 290 \text{ K}$	(A1)

Question	Answer	Marks
3(c)	potential energy (of molecules) is zero	B1
	$U = N \times \frac{1}{2} m \langle c^2 \rangle = 6.0 \times 6.02 \times 10^{23} \times \frac{1}{2} \times 4.7 \times 10^{-26} \times 503^2$ or $U = N \times \left(\frac{3}{2} \right) kT = 6.0 \times 6.02 \times 10^{23} \times \left(\frac{3}{2} \right) \times 1.38 \times 10^{-23} \times 287$ or $U = n \times \left(\frac{3}{2} \right) RT = 6.0 \times \left(\frac{3}{2} \right) \times 8.31 \times 287$	C1
	$U = 21000 \text{ J}$	A1

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